

JFET & MOSFET

Field Effect Transistors in Modern Electronics

Introduction

Definition: A Field Effect Transistor (*FET*) controls current via an electric field at the gate.

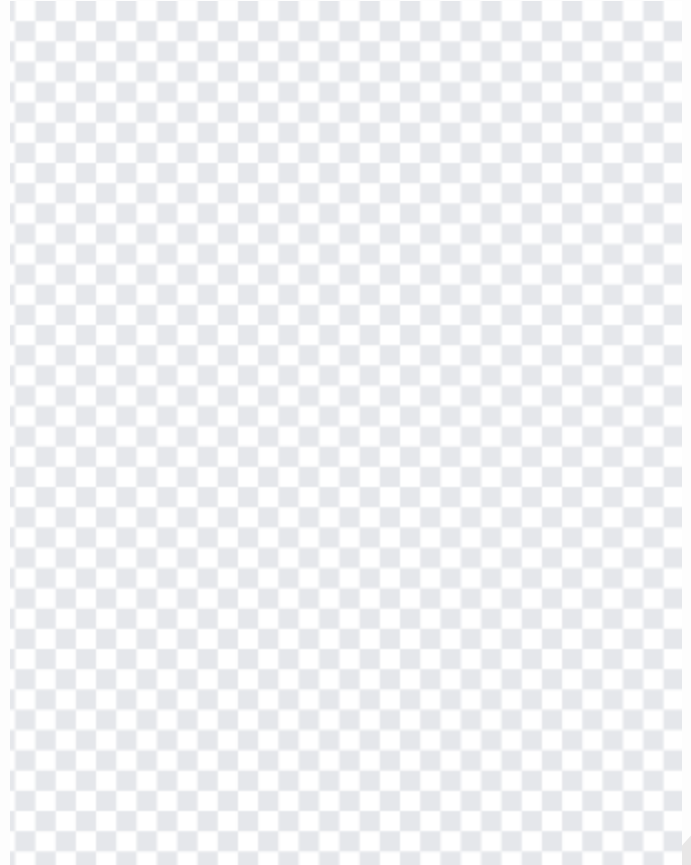
Importance: FETs enable high-input-impedance amplification and efficient switching across analog and digital systems.

Speaker notes: Provide a concise overview. Emphasize FET role in amplifiers, logic, and power management. Mention visuals to follow: symbol, internal cross-section, and I-V characteristic curves.

01

JFET

Fundamentals





Structure & Symbol

Structure: Source, Drain, Gate, and semiconductor channel; *depletion region* controls conduction.

Visuals: include **JFET symbol**, N/P-channel cross-section diagram, and internal structure schematic.

Speaker notes: Point out gate is reverse-biased PN junction (no oxide). Highlight how channel narrowing via depletion region reduces I_d . Show labeled diagrams and symbol for clarity.

Operation & Curves

Operation: Gate voltage modulates depletion width; current is *voltage-controlled*.

Visuals: present transfer (I_d vs V_{gs}) and output (I_d vs V_{ds}) curves; annotate pinch-off and cutoff regions.

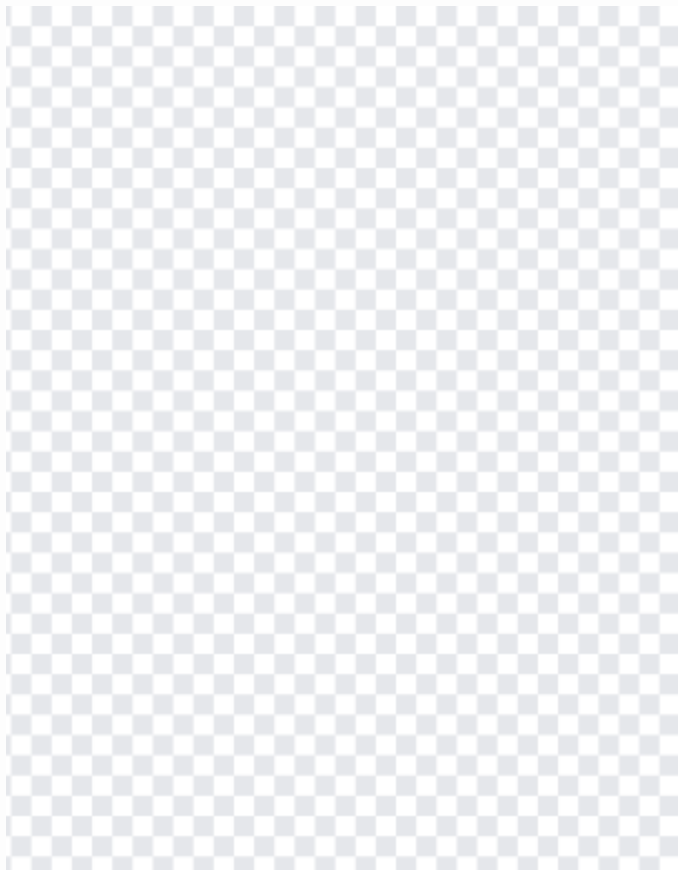
Speaker notes: Explain region behavior—ohmic, pinch-off, saturation. Relate curve features to biasing for amplifier vs switch operation. Recommend overlaying operating points for typical applications.

Applications & Limits

Applications: JFETs used in **low-noise amplifiers**, analog front-ends, and bias-stable circuits.

Limits: Lower input impedance than MOSFETs, limited gate control, and restricted power handling compared to power transistors.

Speaker notes: Show JFET circuit examples and I-V constraints. Include symbol and characteristic curves.



02

MOSFET

Fundamentals

Structure & Types

Structure: Source, Drain, Gate with insulating oxide forming a capacitive gate control.

Types: **Enhancement** and *depletion* modes; N-channel and P-channel variants for complementary designs.

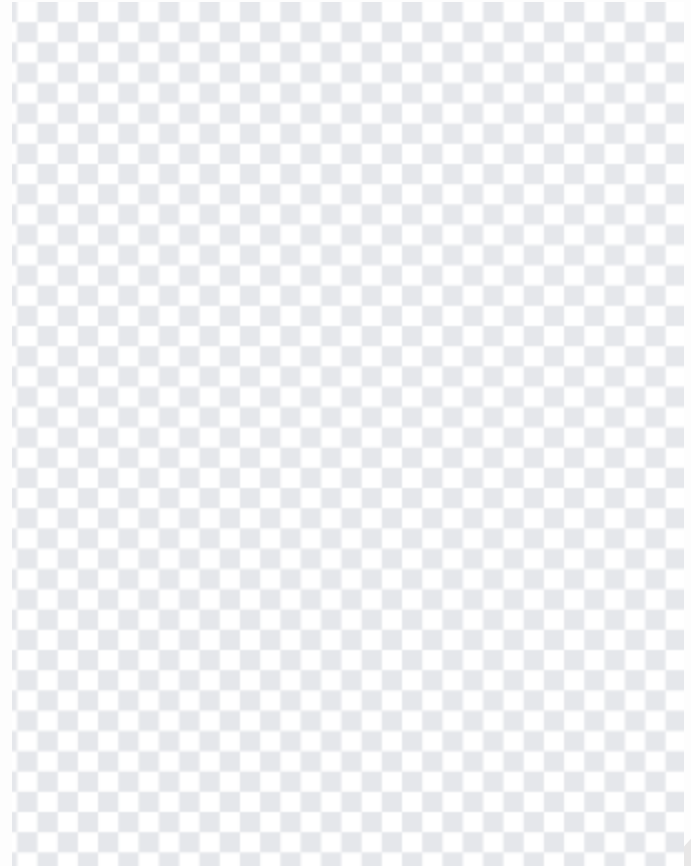
Speaker notes: Display cross-section with oxide layer and gate; compare N vs P channel polarity and symbols.

Applications & Tradeoffs

Applications: Power converters, digital logic, RF amplifiers, and microprocessor power stages.

Tradeoffs: **High input impedance** and fast switching vs gate oxide sensitivity and ESD risk.

Speaker notes: Provide examples: MOSFET in buck converter and CMOS logic. Discuss thermal and reliability considerations.



Conclusions

Summary: JFETs offer simple, low-noise analog performance; MOSFETs deliver high input impedance, fast switching, and scalability for digital and power systems.

Design takeaway: Choose JFET for precision analog; choose MOSFET for power and digital integration.

Speaker notes: Reiterate key differences, show comparative table, and present recommended application map.

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Thanks!

Do you have any questions?

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